

The accountability layer for irreversible agent actions.

The bottleneck for enterprise AI-agent adoption isn't model intelligence — it's **runtime accountability**. IAM and OAuth control *who enters* a system. They cannot prove a **named human authorized a specific irreversible action** — a payment, a vendor-record change, a deletion — **before** an agent executed it. Today that proof is a server log: testimony controlled by the party who acted. EMILIA turns it into evidence.

WHAT IT IS

An **offline-verifiable authorization receipt**: a device-bound human signoff bound to the *exact* action, that anyone can verify with open-source code — **no backend, no vendor trust**. It is native to **MCP**: middleware intercepts an irreversible tool call and requires a valid receipt before release. The thesis in one line: **decision logs are testimony; receipts are evidence**.

SHIPPED & VERIFIABLE — NOT A DECK

- ▶ **ietf Internet-Draft** — draft-schrock-ep-authorization-receipts (citable spec)
- ▶ **Reference verifiers in 3 languages** — JS · Python · Go, cross-language conformance suite
- ▶ **MCP server + middleware** — consent + receipt hook at the tool-call boundary
- ▶ **Live Face-ID demo** — approve a real action, watch a tampered copy get rejected
- ▶ **Formal verification in CI** — 26 TLA+ safety properties (0 errors) · 22 Alloy assertions
- ▶ **One envelope + a profile registry** — EP-ENVELOPE-v1, live at [/.well-known/ep-profiles.json](#)
- ▶ **x402 / AP2 require-receipt rail** — no receipt, no irreversible action
- ▶ **Compliance mappings** — NIST AI RMF · EU AI Act, primitive-by-primitive

Verify a real receipt offline, no account, in ~60 seconds: [npx @emilia-protocol/issue demo](#) · [emiliaprotocol.ai/try](#)

WHY NOW

Security teams are **blocking agent deployment** because an agent can be prompted to act outside its OAuth/IAM gates, and "a human approved this" can't be proven after the fact. NIST AI RMF and federal agent-oversight guidance are moving "*who approved this?*" from best practice toward requirement — first in government and finance, where a missing approval is a statutory problem, not just a bad day.

DISTRIBUTION → REVENUE

Open standard wins the chokepoint (MCP SEP + the require-receipt rail); the managed layer — issuance, registry, audit (GovGuard / FinGuard) — is the revenue.

HONEST STATUS

Pre-revenue, pre-customer. An open, Apache-2.0 standard with a formally-verified reference stack. Raising a **seed** to convert that into requiredness and a technical co-founder.

EMILIA Protocol, Inc. — U.S. small entity, work by U.S. persons. Apache-2.0.
Optional: receipts can be Merkle-anchored for public timestamping; the core needs no blockchain.

[Iman Schrock](#), Founder
team@emiliaprotocol.ai · [emiliaprotocol.ai](#)
github.com/emiliaprotocol/emilia-protocol